

Supplemental Material for ‘How Parliaments Polarize: Legislative Replacement and Adaptation in Chile’

Under review

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SM1: Dimensionality

The substantive quantities of interest (distance-based measures of extremity and chamber-level dispersion over time) are meaningful only relative to the geometry of the underlying preference space. If roll-call voting were genuinely multi-dimensional in a stable way, a one-dimensional summary could distort distances and misstate polarization trends. For that reason, it is standard in roll-call research to treat one-dimensional dynamics as the baseline only after checking whether an additional dimension contributes explanatory power (Poole and Rosenthal 1997).

We assess dimensionality using a two-dimensional DW-NOMINATE implementation and extract eigenvalues for each period’s roll-call vote matrix. Each estimation uses either the right-wing or the left-wing deputy mentioned in the main text as the anchor. Table S1 reports the leading eigenvalues λ_1, λ_2 , their ratio λ_1/λ_2 , and implied variance shares $\lambda_k/(\lambda_1 + \lambda_2)$ by period. When λ_1/λ_2 is large and the first-dimension share is high, the empirical structure of voting is predominantly one-dimensional, and focusing the dynamic analysis on the first dimension is appropriate for measuring polarization along the main cleavage.

In this dataset, across all periods, λ_1 is at least fourteen times larger than λ_2 , implying that the first dimension accounts for more than 93% of the overall dispersion in the roll-call data within each period. Given the dominance of the first dimension, the added complexity of a dynamic two-dimensional specification is unlikely to change inference about the main polarization trend, while potentially reducing interpretability. Finally, anchor choice affects only the orientation (sign) of the recovered scale; it does not affect eigenvalue-based dimensionality diagnostics, which are invariant to sign reversals.

Table 1: DW-NOMINATE 2D dimensionality diagnostics by period: eigenvalues, dominance ratio (λ_1/λ_2), implied variance shares, and anchors used.

period	λ_1	λ_2	λ_1/λ_2	share dim 1	share dim 2	right-wing anchor	left-wing anchor
1	12.311	0.770	15.978	0.941	0.059	✓	✓
2	14.475	0.716	20.219	0.953	0.047	✓	✓
3	13.896	0.586	23.731	0.960	0.040	✓	✓
4	10.111	0.715	14.137	0.934	0.066		✓
5	19.441	1.017	19.121	0.950	0.050	✓	✓
6	18.106	0.644	28.110	0.966	0.034	✓	

SM2: Robustness to Non-Voting (Absences and Abstentions)

Non-voting (absences and abstentions) can in principle distort ideal-point estimation if it is systematically related to legislators’ latent preferences or changes over time. We therefore re-estimate the baseline dynamic IRT model (three-anchor specification) after excluding legislators whose non-vote rate within any served period exceeds alternative thresholds (25%, 35%, and 50%), where “non-vote” is defined as any response other than Yea (“Afirmativo”) or Nay (“En Contra”). Numbers and proportions are in Table S2. The descriptive diagnostics show that absences decline over time. The robustness results in Table S3 show very high correspondence at the legislator-period level (Pearson correlations of 0.996–0.999) and strong correspondence for the six-point polarization series (Pearson correlations of 0.939–0.993). The most restrictive 25% threshold removes 216 of 430 legislators and produces the largest short-run deviations, but the completed 2022–26 term remains the most polarized under every threshold. These results imply that the long-run increase and the final-period maximum are not artifacts of participation patterns or a small number of low-participation legislators.

Table 2: Legislator exclusions under alternative non-vote thresholds (non-vote = any response other than {Afirmativo, En Contra}).

Threshold	Excluded (n)	Excluded (%)	Kept (n)	Universe (n)
25%	216	50.23	214	430
35%	118	27.44	312	430
50%	48	11.16	382	430

Table 3: Robustness to non-votes: correlations and linear fits of filtered estimates on the full baseline (three-anchor dynIRT).

Threshold	Outcome	N pairs	Pearson r	Spearman	Intercept	Slope	R ²
25%	Ideal points (DiputadoId × Período)	361	0.9956	0.9913	-0.0079	0.9648	0.9911
35%	Ideal points (DiputadoId × Período)	587	0.9994	0.9984	-0.0401	1.0039	0.9989
50%	Ideal points (DiputadoId × Período)	736	0.9994	0.9987	-0.0212	1.0042	0.9989
25%	Polarization series (period means)	6	0.9394	0.6571	0.0278	1.1143	0.8825
35%	Polarization series (period means)	6	0.9812	0.8286	0.0187	1.0848	0.9627
50%	Polarization series (period means)	6	0.9925	0.9429	0.0335	1.0064	0.9851

SM3. Robustness to anchor choice

The main text relies on a multi-anchor solution to fix the orientation of the ideological scale. In Table S4 we document (i) which legislators were used as anchors, (ii) their availability and voting participation, and (iii) the extent to which alternative single-anchor solutions yield nearly identical ideal-point estimates (up to sign reversals). Because anchor choice can flip the orientation of the recovered scale, we align the sign of each single-anchor estimate to the baseline multi-anchor estimate before computing correlations and generating pairwise scatterplots.

Table 4: Anchor candidates used in single-anchor estimations (and baseline triad): availability across periods and overall voting participation.

DiputadoId	Name	Anchor type	Periods present (n)	Vote rate overall (%)
917	Gaston Von Muhlenbrock Zamora	right-wing	5	89.9
865	Patricio Melero Abaroa	right-wing	5	80.5
811	Ramon Barros Montero	right-wing	5	77.1
850	Javier Hernandez Hernandez	right-wing	5	76.7
908	Ignacio Urrutia Bonilla	right-wing	5	74.9
879	Jose Miguel Ortiz Novoa	center- moderate	5	95.7

DiputadoId	Name	Anchor type	Periods present (n)	Vote rate overall (%)
897	Alejandra Sepulveda Orbenes	left-wing	5	85.5
855	Carlos Abel Jarpa Wevar	left-wing	5	79.5
837	Fidel Espinoza Sandoval	left-wing	5	70.4

Although the estimates are not identical, their correlations are all close to 1, as shown in Figure S1 below.

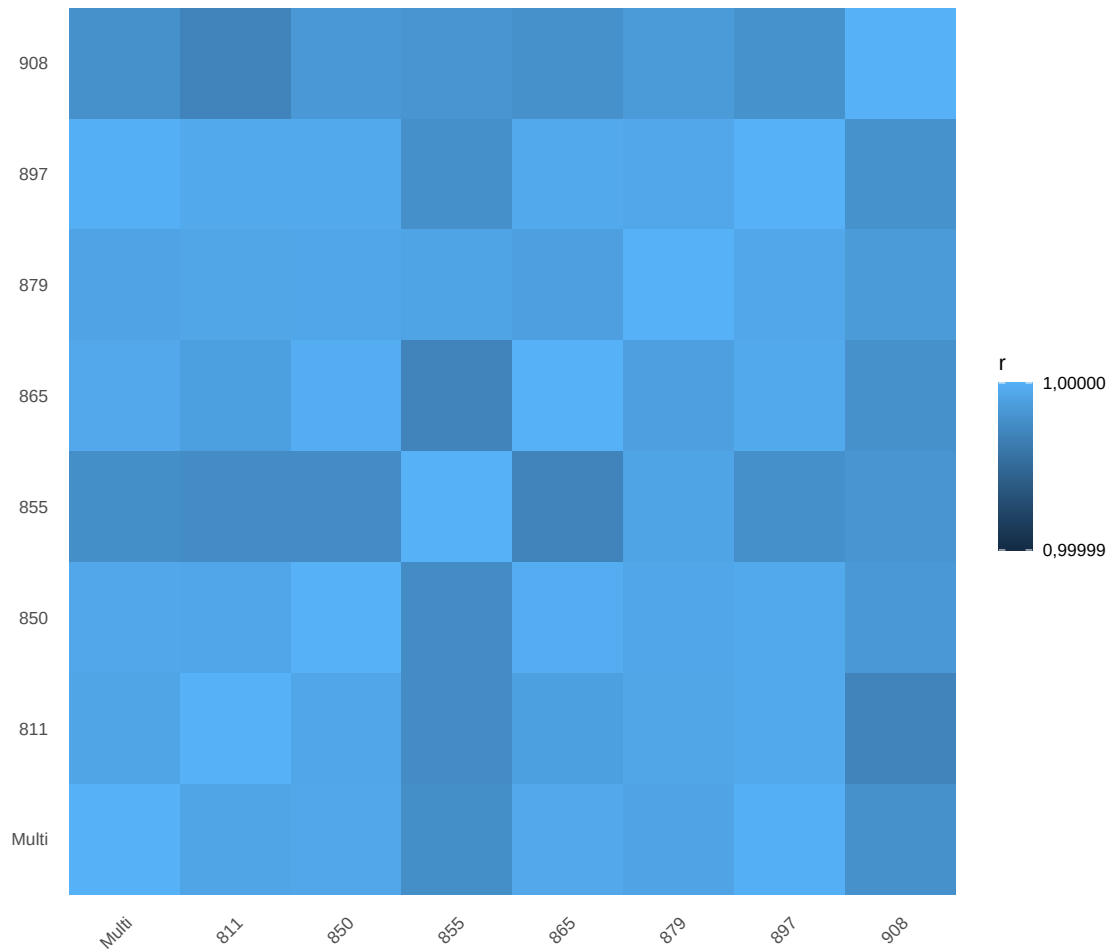


Figure 1: Correlation heatmap (sign-aligned). Color scale restricted to $[0.90, 1.00]$.

Table 5: Legislator-level ideal-point uncertainty by period (CI width summary).

period_lab	median	p25	p75	min	max
2002-06	0.074	0.074	0.075	0.074	0.075
2006-10	0.080	0.080	0.080	0.080	0.080
2010-14	0.060	0.060	0.060	0.060	0.060
2014-18	0.053	0.053	0.053	0.053	0.053
2018-22	0.059	0.059	0.059	0.059	0.059
2022-26	0.049	0.049	0.049	0.049	0.049

SM4. Estimation uncertainty and uncertainty propagation

To propagate legislator–period ideal-point uncertainty into period-level polarization, we use the 95% intervals reported by *dynIRT* (package *emIRT* in R). Treating $[x_{it}^{low}, x_{it}^{high}]$ as an approximate 95% interval around the point estimate $\hat{x}_{it}$, we recover $\hat{\sigma}_{it} \approx (x_{it}^{high} - x_{it}^{low}) / (2 \cdot 1.96)$ under a normal approximation. We then simulate $x_{it}^{(b)} \sim \mathcal{N}(\hat{x}_{it}, \hat{\sigma}_{it}^2)$ for $b = 1, \dots, B$, recompute period-level polarization for each draw, and report uncertainty bands as the 2.5th and 97.5th percentiles of the draw-level polarization distribution.

Table S5 (median and IQR of interval widths by period) shows that ideal-point uncertainty is highly homogeneous within legislatures: variation in interval widths across legislators within a period is negligible, so any differences in precision arise primarily across periods rather than from a small subset of unusually imprecise legislators. Consistent with this pattern, the propagated uncertainty bands for polarization are narrow and do not affect the ordering of periods or the substantive interpretation of the polarization trajectory.

The uncertainty intervals in Table S6 quantify how ideal-point uncertainty propagates into period-level polarization summaries. For each period, we take the estimated legislator ideal points $\hat{x}_{it}$ and recover an approximate standard deviation from the reported 95% interval $[x_{it}^{low}, x_{it}^{high}]$. We then draw Monte Carlo realizations $x_{it}^{(b)} \sim \mathcal{N}(\hat{x}_{it}, \hat{\sigma}_{it}^2)$ and recompute two polarization measures in each draw: mean absolute ideal point (*abs*) and mean squared ideal point (*sq*). The table reports the Monte Carlo mean (“Estimate”) and the central 95% interval (2.5%–97.5%) of each measure’s sampling distribution induced solely by ideal-point uncertainty.

Two patterns are evident. First, the propagated intervals are very narrow in every period and for every measure (typically a few thousandths wide), indicating that the main time-series movements in polarization are orders of magnitude larger than the uncertainty introduced by estimation error in individual ideal points. Second, the time-profile of each measure remains unchanged once uncertainty is propagated: polarization rises across the six complete periods and reaches its maximum in 2022–26 under both absolute and squared positions.

Table 6: Uncertainty-propagated polarization intervals by period (Monte Carlo propagation from ideal-point uncertainty).

Period	Measure	Estimate	Low (2.5%)	High (97.5%)
2002-06	abs	0,4792	0,4759	0,4825
2002-06	sq	0,3546	0,3506	0,3587
2006-10	abs	0,5167	0,5129	0,5203
2006-10	sq	0,3610	0,3565	0,3652
2010-14	abs	0,5746	0,5719	0,5774
2010-14	sq	0,4652	0,4614	0,4689
2014-18	abs	0,5963	0,5939	0,5987
2014-18	sq	0,5709	0,5673	0,5747
2018-22	abs	0,7213	0,7190	0,7236
2018-22	sq	0,6651	0,6614	0,6688
2022-26	abs	0,8435	0,8416	0,8454
2022-26	sq	0,9002	0,8967	0,9038

A separate measurement choice concerns the reference point used to summarize ideological extremity. We do not rely on alternative polarization measures defined as average absolute distance from the period-specific mean or median. Such measures are informative about dispersion around the contemporaneous center, but they anchor polarization to reference points that move endogenously with the distribution itself. Over long horizons, this can obscure changes in ideological positioning when the entire distribution shifts, because dispersion around the contemporaneous center may remain stable even as legislators move farther from a fixed longitudinal benchmark. Because the objective of the analysis is to track polarization relative to a stable reference point over time, these moving-anchor measures are not pursued here.

SM5. Additional Implementations and Diagnostics (Model Stability Across Designs)

To assess whether the upward trend documented in the main text depends on the specific polarization metric, we plot it again in Figure S2 together with an alternative operationalization from the same ideal-point estimates and normalize both series to the first period (2002–06 = 1). The left panel reports mean absolute ideological position, $\mathbb{E}(|x|)$ (*abs*), which tracks average dispersion from the reference point in a linear way; the right panel reports mean squared ideological position, $\mathbb{E}(x^2)$ (*sq*), which places disproportionate weight on more extreme legislators. Both measures reproduce the same qualitative dynamics as the baseline series in the main text: polarization rises across the observation window and reaches its maximum in the completed 2022–26 term. Relative to 2002–06, the 2022–26 indices are 1.76 for the absolute

measure and 2.54 for the quadratic measure. The stronger increase under the quadratic measure indicates that the overall rise is not driven solely by moderate dispersion, but also by a growing contribution of legislators located farther in the tails of the ideological distribution.

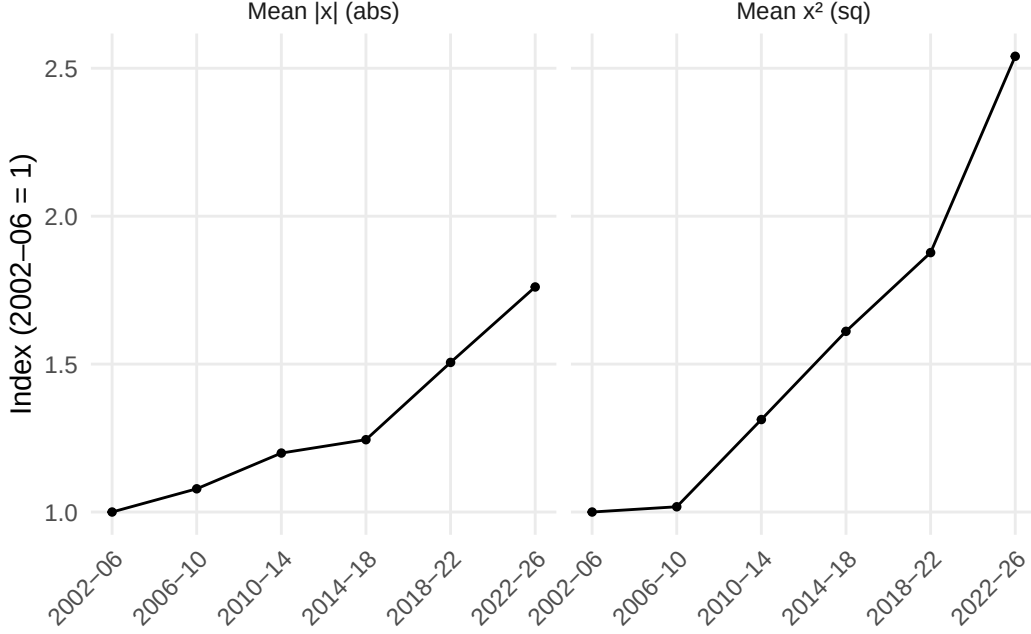


Figure 2: Trends in polarization metrics

Within-legislator adaptation specification

Model 6 evaluates whether legislators become more or less extreme relative to the parliamentary center as their careers unfold. The outcome, $\Delta d_{i,t}$, measures the change since entry in each legislator's absolute distance from the contemporaneous median of incumbents. Positive values therefore indicate increasing relative extremity, whereas negative values indicate movement closer to that benchmark, regardless of whether the legislator is located on the left or the right. The alternative specifications reported below replace the incumbent median with the corresponding mean or with full-chamber benchmarks, while keeping the same legislator panel and HC1-clustered inference.

Table 7: Model 6 linear specifications: change in relative extremity since entry.

Measure	Benchmark	Statistic	Estimate	Clustered SE	95% CI	p-value	N	Legislators
Change in distance since entry		Median	0.006	0.017	[-0.028, 0.039]	0.743	609	230
		Mean	0.008	0.011	[-0.013, 0.029]	0.470	609	230
		Median	-0.010	0.018	[-0.045, 0.025]	0.575	609	230

Table 7: Model 6 linear specifications: change in relative extremity since entry. (*continued*)

Measure	Full chamber		Statistic	Estimate	Clustered SE	95% CI	p-value	N	Legislators
	Benchmark								
Outward change since entry			Mean	-0.006	0.011	[-0.028, 0.016]	0.613	609	230
Change in distance since entry	Incumbents		Median	-0.003	0.016	[-0.033, 0.028]	0.848	355	148
			Mean	0.001	0.018	[-0.034, 0.036]	0.945	355	148
Outward change since entry			Median	-0.030	0.018	[-0.066, 0.005]	0.095	355	148
			Mean	-0.010	0.019	[-0.047, 0.026]	0.574	355	148

To assess whether the linear specification masks nonlinear changes over legislative careers, we also estimate Model 6 nonparametrically by τ , the number of legislative periods elapsed since entry. The primary specification uses the incumbent median as the benchmark, while the full-chamber median provides a robustness comparison. Because $\tau = 0$ corresponds mechanically to the entry period, it serves only as the reference category; the substantive estimates begin with subsequent periods in office.

Table S8: Model 6 nonparametric estimates by τ .

Benchmark	tau	Estimate	Clustered SE	95% CI	p-value	N observations	N legislators
Incumbent median	0	-0	8.057e-19	[-1.579e-18, 1.579e-18]	1	148	148
Incumbent median	1	-0.01344	0.02438	[-0.06122, 0.03433]	0.5817	143	143
Incumbent median	2	-0.01738	0.04037	[-0.09651, 0.06174]	0.667	51	51
Incumbent median	3	0.01101	0.08585	[-0.1573, 0.1793]	0.898	11	11
Incumbent median	4	0.2376	0.02736	[0.184, 0.2912]	1.486e-16	2	2
Full-chamber median	0	-0	4.895e-19	[-9.595e-19, 9.595e-19]	1	230	230
Full-chamber median	1	-0.005315	0.02457	[-0.05346, 0.04284]	0.8288	221	221
Full-chamber median	2	0.01012	0.0359	[-0.06025, 0.08048]	0.7782	98	98
Full-chamber median	3	-0.05158	0.07436	[-0.1973, 0.09416]	0.4882	37	37
Full-chamber median	4	0.1026	0.1089	[-0.1109, 0.3161]	0.3466	21	21
Full-chamber median	5	0.3817	0.3036	[-0.2134, 0.9767]	0.2092	2	2

The nonparametric estimates show little indication of a systematic increase in extremity over legislators' careers. At longer horizons, some coefficients turn positive, but these estimates are based on very small numbers of legislators: in the full-chamber specification, only 21 legislators contribute at $\tau = 4$ and two at $\tau = 5$, while the incumbent-benchmark specification includes only two observations at $\tau = 4$ and none at $\tau = 5$. These late-horizon estimates are therefore too sparse to support an inference of a general adaptation process. Taken together, the nonparametric results are consistent with the linear specification and provide no systematic evidence of increasing relative extremity after entry.

On the stability of key coefficients in the regressions

Table S9 assesses whether the main level-model results remain stable across alternative samples and outcome definitions. The comparison covers the legislative-period trend, the entrant-incumbent difference, and the post-reform terms from Models 1–5, using both the absolute-distance and squared-ideology measures. The estimates are grouped into four design families: the full sample (A), subsamples by bill origin (B), subsamples by legislative stage (C), and 100 balanced samples that hold the number of roll calls constant across periods (D). For each combination, the table reports how often the 95% confidence interval excludes zero. Model 6 is considered separately because it evaluates within-legislator change rather than outcome levels.

The overall pattern is straightforward. The legislative-period trend is the most stable result, remaining statistically distinguishable from zero across all designs and both outcome measures. The entrant-incumbent difference is also highly consistent, although it becomes less precise in some analyses by legislative stage, where splitting the data reduces information and may expose greater heterogeneity across procedural settings. The post-reform terms are considerably less stable and do not display a comparable pattern. Overall, these results reinforce the interpretation that the long-run rise in polarization and the systematic difference between entrants and incumbents are robust features of the data, whereas evidence of a discrete post-reform shift is much weaker.

Table 9: Share of subsample coefficient estimates whose 95% CI excludes or crosses zero, by term, outcome definition, and design.

Term	DV	Design	N	Excludes 0	Crosses 0
Trend	abs	A	3	100.0%	0.0%
Trend	abs	B	6	100.0%	0.0%
Trend	abs	C	12	100.0%	0.0%
Trend	abs	D	3	100.0%	0.0%
Trend	sq_ideology	A	1	100.0%	0.0%
Trend	sq_ideology	B	2	100.0%	0.0%
Trend	sq_ideology	C	4	100.0%	0.0%
Trend	sq_ideology	D	1	100.0%	0.0%

Term	DV	Design	N	Excludes 0	Crosses 0
Replacement	abs	A	3	100.0%	0.0%
Replacement	abs	B	6	100.0%	0.0%
Replacement	abs	C	12	75.0%	25.0%
Replacement	abs	D	3	100.0%	0.0%
Replacement	sq_ideology	A	1	100.0%	0.0%
Replacement	sq_ideology	B	2	100.0%	0.0%
Replacement	sq_ideology	C	4	50.0%	50.0%
Replacement	sq_ideology	D	1	100.0%	0.0%
Post-reform	abs	A	3	0.0%	100.0%
Post-reform	abs	B	6	33.3%	66.7%
Post-reform	abs	C	12	58.3%	41.7%
Post-reform	abs	D	3	0.0%	100.0%
Post-reform	sq_ideology	A	1	0.0%	100.0%
Post-reform	sq_ideology	B	2	0.0%	100.0%
Post-reform	sq_ideology	C	4	25.0%	75.0%
Post-reform	sq_ideology	D	1	0.0%	100.0%

Tail dynamics of ideological polarization

Figure S3 examines whether the long-run increase in polarization is also visible in the upper tail of the ideological distribution. Average dispersion captures broad shifts in legislative positioning, while tail-based indicators show whether those changes extend to legislators located farthest from the ideological reference point. All series are indexed to the first period (2002–06 = 1).

The first two panels track the 90th and 95th percentiles of absolute ideological position. Both increase substantially over the observation window and reach their highest levels in 2022–26. Relative to 2002–06, the p90 index rises to 1.50 and the p95 index to 1.71. The long-run increase documented in the main analysis therefore extends well into the upper tail of the distribution rather than being confined to shifts around its center.

The third panel asks a different question: whether overall quadratic dispersion becomes increasingly concentrated among the most extreme legislators. It reports the share of total $\sum x^2$ accounted for by the top 10% of legislators ranked by $|x|$, again indexed to 2002–06. The series is non-monotonic: it rises modestly in 2010–14, reaches its minimum in 2018–22, and returns almost exactly to its baseline level in 2022–26. Thus, greater high-percentile extremity does not coincide with a sustained increase in the concentration of polarization among a small group of legislators. The long-run rise appears instead to involve a broader outward shift in the ideological distribution.

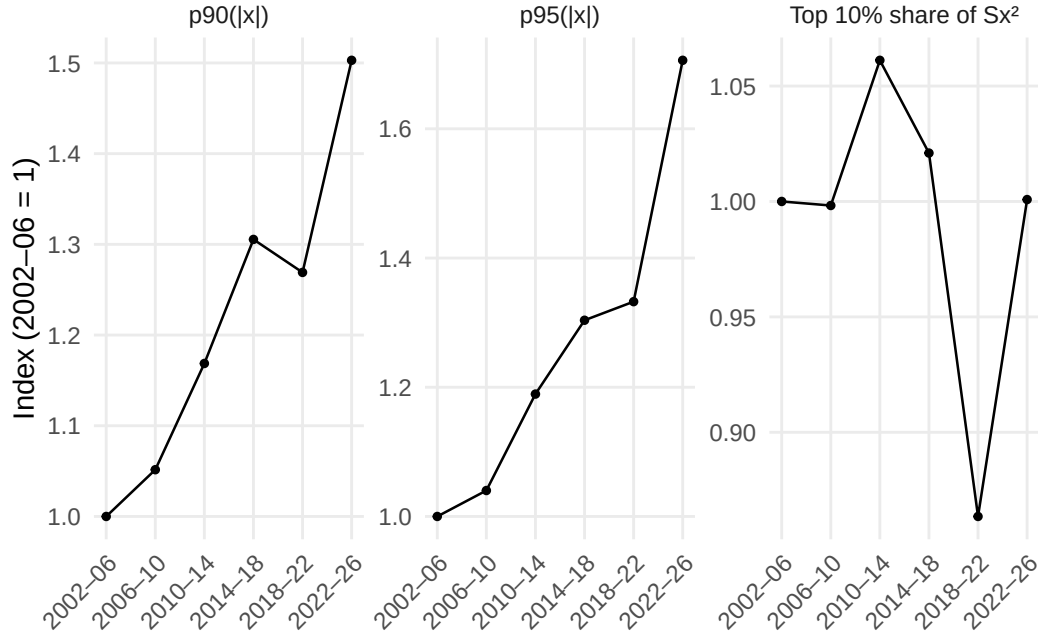


Figure 3: Upper tails of the ideological distributions

SM6 — Exact replacement-only accounting and cohort profiles

SM6 evaluates how much of the long-run increase in polarization can be reproduced by changes in chamber composition alone. Using the same dynamically anchored legislator-period ideal points as the main models, each legislator is held at the level of extremity observed when they first enter the dataset, while the actual membership of the Chamber is allowed to evolve across periods. This yields an exact accounting decomposition in which observed polarization is separated into an entry-stock component and the average within-legislator deviation from entry extremity. The exercise is descriptive rather than causal. Because some legislators observed in 2002–06 may have entered the Chamber before the beginning of the study window, the first cohort is potentially left-censored.

Table 10: Exact replacement-only decomposition of mean absolute anchored ideology by period.

Period	Observed	Entry-stock / replacement-only	Within-legislator deviation	Identity difference
2002-06	0.4790	0.4790	0.0000	0.0e+00

Period	Observed	Entry-stock / replacement-only	Within-legislator deviation	Identity difference
2006-10	0.5166	0.5204	-0.0038	-1.3e-18
2010-14	0.5745	0.5678	0.0067	2.5e-17
2014-18	0.5962	0.6541	-0.0579	-1.4e-17
2018-22	0.7213	0.6961	0.0252	-2.4e-17
2022-26	0.8435	0.8454	-0.0019	2.5e-17

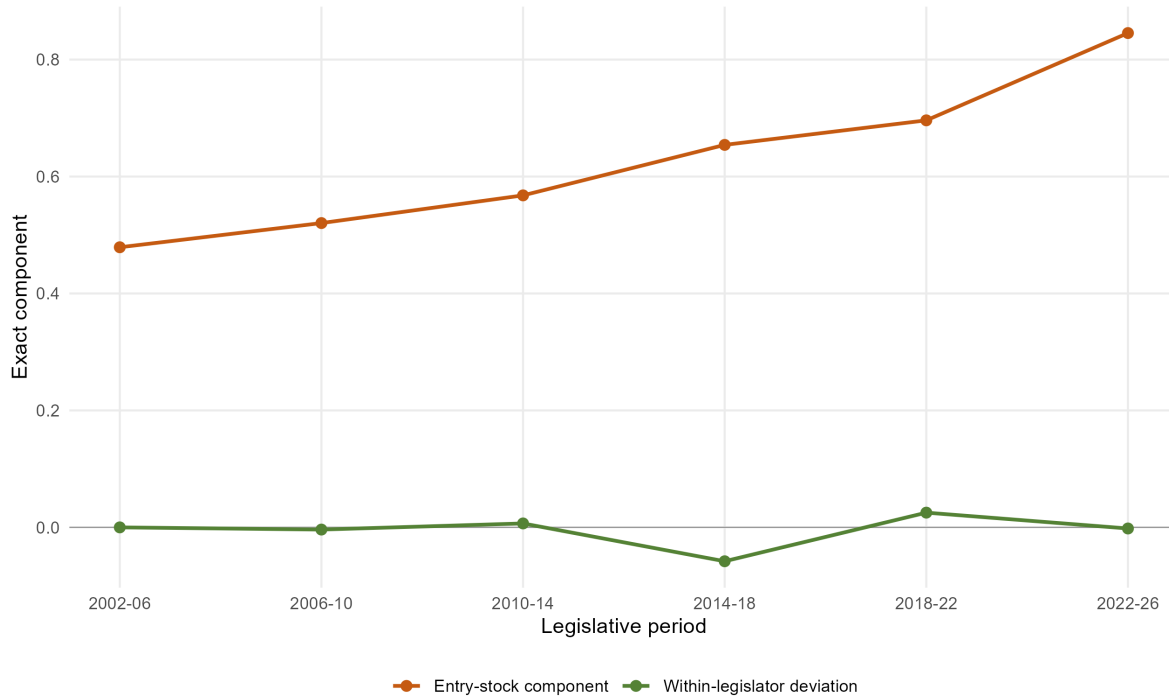


Figure 4: Entry-stock and aggregate within-legislator components by period.

Entry extremity rises across successive first-observed cohorts, from 0.479 in 2002–06 to 0.900 in 2022–26, with intermediate cohort means of 0.537, 0.641, 0.713, and 0.759. This progression is consistent with cumulative replacement, although the analysis does not identify why later cohorts enter at increasingly extreme positions.

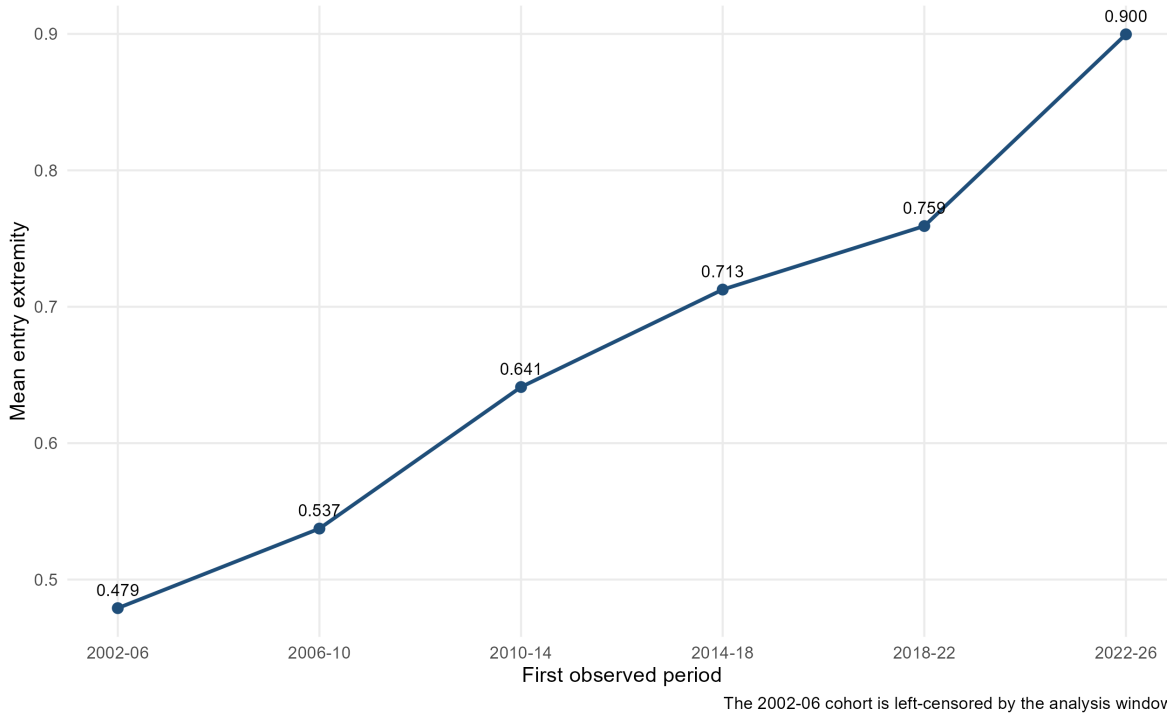


Figure 5: Mean entry extremity by first-observed cohort.

Table 11: Quadratic-measure robustness, 2002–06 to 2022–26.

Observed increase	Replacement-only increase	Accounting ratio	Within component
0.5457	0.5504	100.85%	-0.0047

The quadratic specification leads to the same substantive conclusion. Observed polarization increases by 0.5457, while the replacement-only series increases by 0.5504, yielding a descriptive accounting ratio of 100.85%. The final within-legislator component is again effectively zero (-0.0047), reinforcing the conclusion that the long-run increase is accounted for almost entirely by compositional replacement.

References

Poole, Keith T., and Howard Rosenthal. 1997. *Congress: A Political-Economic History of Roll Call Voting*. New York: Oxford University Press.